



LESSON - 10

SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

CASE STATEMENT

The CASE statement is used for conditional logic, similar to IF-ELSE in programming languages.

Syntax:

CASE case_value

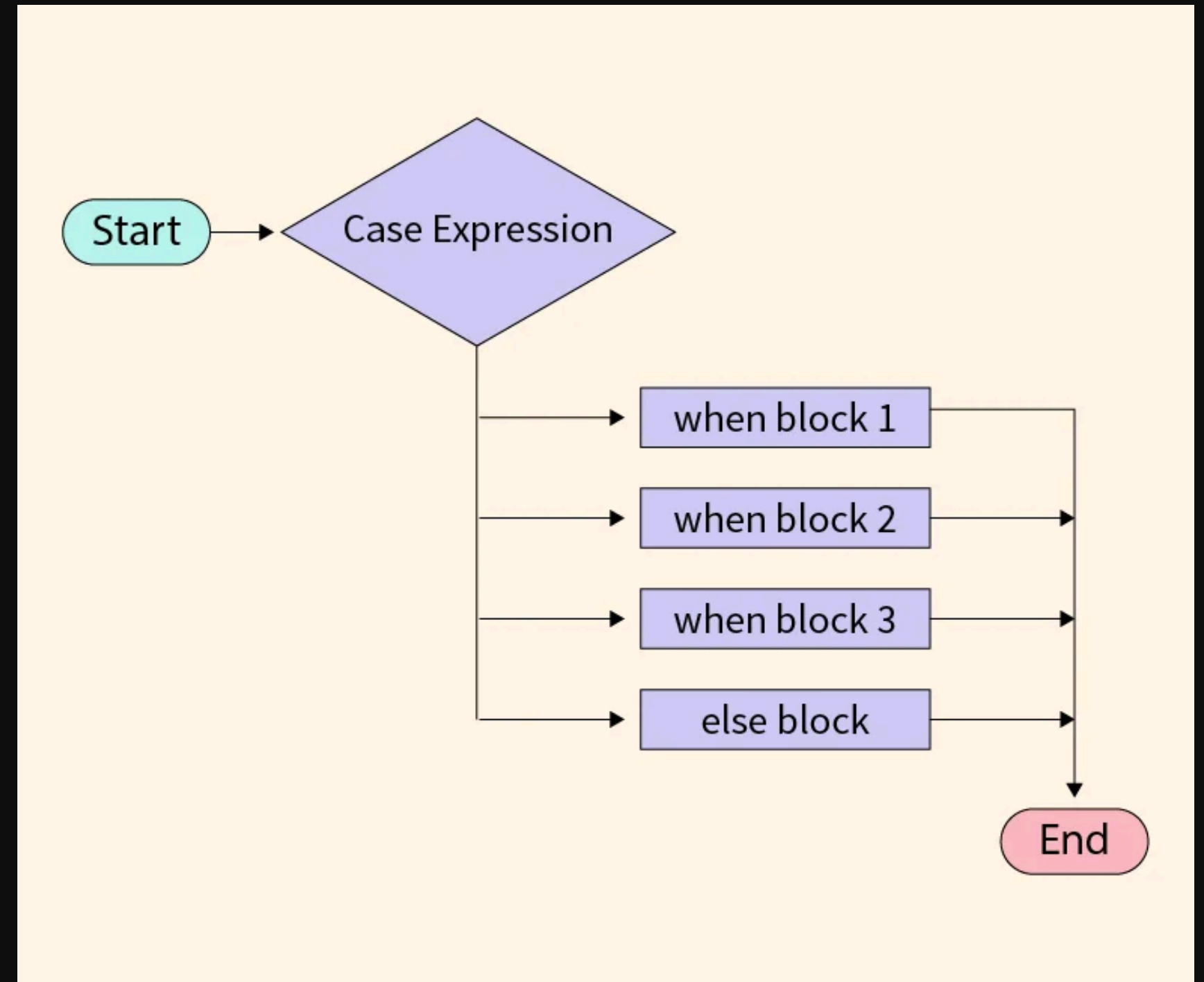
WHEN condition THEN result1

WHEN condition THEN result2

...

Else result

END CASE;



COALESCE

Returns the first non-NULL value from a list of expressions.

Syntax:

```
SELECT
    employee_name,
    COALESCE(phone_number, 'Not Provided') AS contact FROM
employee;
```

```
SELECT
    employee_name,
    COALESCE(email, phone_number, 'No Contact Info') AS contact_info
FROM employee;
```

NULLIF

Returns NULL if two expressions are equal; otherwise, it returns the first expression.

Syntax:

SELECT

sales, sales / NULLIF(target, 0) AS performance

FROM employee;

SELECT

employee_name, NULLIF(bonus, 0) AS bonus_received

FROM employee;